

Project: Telecom Customer Churn Prediction

Objective:

Develop a machine learning model to predict whether a telecom customer is likely to churn (leave the service) based on their usage patterns, account details, and subscription type.

Dataset:

telco_customer_churn.csv

https://drive.google.com/file/d/1n3SNdPX4QFelmdPHS9AkkAaafgUAo24/view?usp=drive_link

Target Column:

Churn (Yes = customer will churn, No = customer stays)

Project Workflow

1. EDA & Data Cleaning

- Explore customer demographics (gender, senior citizen, partner, dependents).
- Examine account details (tenure, contract type, payment method, paperless billing).
- Analyze service features (InternetService, StreamingTV, PhoneService, etc.) and their relationship with churn.
- Handle missing values
- Encode categorical variables
- Scale numerical features
- Check for class imbalance in churn distribution.

2. Build a Training Pipeline

- Use `sklearn.pipeline.Pipeline` to combine preprocessing (scaling, encoding) and the ML model.
- Start with simple models (Logistic Regression, Decision Trees) and then test more advanced ones (Random Forest, XGBoost).

3. Train & Evaluate

- Split dataset into training & test sets.
- Train multiple models and compare their performance using metrics:
 - Accuracy
 - Precision
 - Recall
 - F1 Score
 - ROC-AUC
- Use hyperparameter tuning (GridSearchCV / RandomizedSearchCV)

4. Save the Model

- Export the final trained pipeline to a .pkl file using pickle.

5. Inference Script

- Write a script that loads the saved model.
- Take sample input (customer's subscription details) from terminal or a JSON file.
- Predict churn (Yes/No) and print a human-readable result (e.g., 'This customer is at risk of leaving').

Key Considerations

GitHub Project Setup

- Create a GitHub repository with project name Telecom-Customer-Churn.
- Use clear commit messages (e.g., 'Completed task Telecom Customer Churn prediction').
- Keep Jupyter notebooks in a notebooks/ folder.

Code Quality

- Use meaningful variable names.
- Add comments explaining data preprocessing and model logic.
- Structure scripts in reusable functions instead of writing everything in one cell.

Suggested Folder Structure

```
Telecom-Customer-Churn/  
| — data/          # dataset  
| — notebooks/     # Jupyter notebooks for EDA & experiments and main file  
| — models/        # saved model files (.pkl)  
| — README.md      # project description  
| — requirements.txt # dependencies  
| — .gitignore     # ignore dataset & unnecessary files
```

Tip:

Focus more on cleaning and preprocessing, since this dataset is messy and will test your understanding of handling categorical variables, missing values, and imbalanced targets.